Latent Causal Infeasibility for Long-Horizon TAMP

Methodology specification v1.5 | Initial-scene repair without trajectory inputs

16 October 2026

Reading contract and revision decision

Primary objective. Given the initial observed scene, supplied object geometry, robot state, legal resting destinations and a parameterized action/goal, predict a small set of obstacle relocations that enables the intended action. The learned proposal module receives no target trajectory, swept-volume representation or collision-oracle output. IK, relocation ordering, robot motion planning and target-action evaluation remain external components.

Primary inference protocol. Predict a complete endpoint assignment from the initial scene. An external relocation planner searches a legal order and motions for that assignment. Commit to a repair before receiving any target-planning result. Execute the repair, then invoke the target-action planner/evaluator from the achieved state. Do not test or replan the target action between relocations. Observe execution deviations for validation or abort; do not silently use them to update the proposal in the primary experiment.

This is a revised research contract, not evidence that the revised system works. It supersedes specification v1.4 for new primary experiments. Implementation Guide v1.7 is its companion. The old master document is historical motivation, not a competing instruction source. The next authorized development gate is **G3.5: objective migration and case analysis**, before G4 data scaling or G5 learning.

What changes, precisely

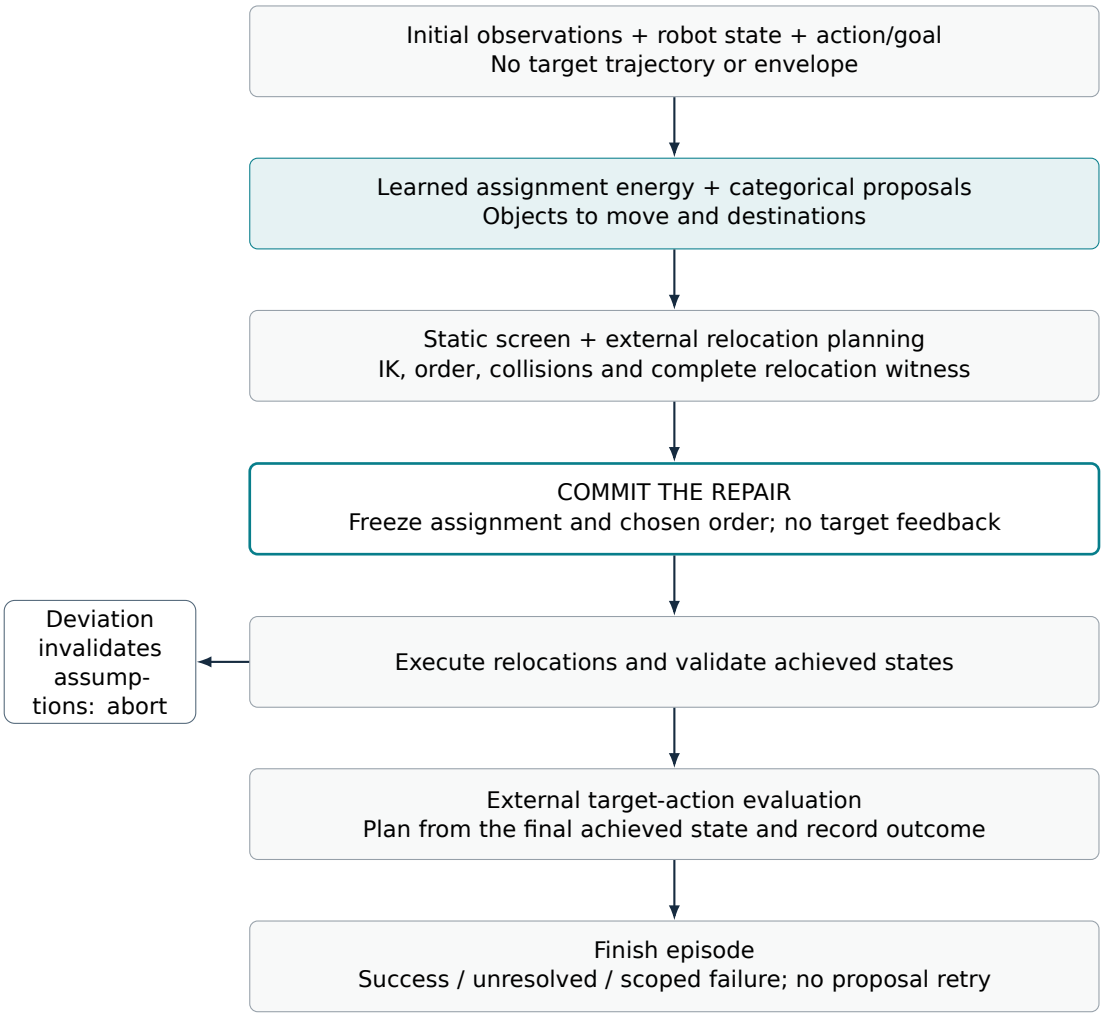
Previous contract	New primary contract
Restore one validated, frozen target motion	Enable the declared action through an allowed motion
Envelope supplied to the learner	Initial scene, robot and action/goal supplied; no trajectory channel
Per-object action-clear factors define target clearance	Static endpoint factors plus a learned whole-assignment success score
Frozen-motion replay filters proposed repairs	Target evaluation occurs after repair commitment; no feedback to selection
Direct blocker count can prove a frozen-route minimum	No such lower bound for the new objective
Reobservation may recompute energies	Primary proposal remains tied to the initial observation

Preserved. Full articulated collision checking, supported finite endpoints, stable object identities, move-once episodes, achieved-state execution, conservative verification, uncertainty labels, reproducible splits and analytic controls.

Archived. G0-G3 remain valid at their original scope. Report 0006 describes 74 passing tests and fixed-motion executable witnesses; this revision does not independently rerun them or reinterpret those witnesses as action-level minima. Frozen-motion experiments remain available as `FROZEN_CONTROL`.

System boundary: what is learned and what is executed

The path from initial observation to committed repair is the method being studied. The final target-action outcome evaluates that decision. The offline teacher can use richer evidence to create labels, but its planner outputs are not online inputs.



Offline supervision is separate. A teacher may evaluate many assignments and target motions to find witnesses or certify negatives. The resulting labels supervise the model; reference paths, outcomes and search traces remain privileged. A primary proposal cannot query that teacher.

What one-shot means here. The complete high-level assignment uses the initial observation. Sampling many hypothetical assignments and external relocation-order search are allowed before commitment. Reobserving execution for validation is allowed afterward. A target-guided change to the selected assignment is a different protocol.

1. Scope, inputs and output

One query concerns one active target object and one intended action. A sequence of household actions is a separate integration experiment. The initial learned task is supported transport of a target to a declared goal; articulated opening cases require their own object-joint/contact implementation before being reported as robot results.

Let S_0 denote the complete simulator or physical state; o_0 the available initial observation; a the parameterized action; g its goal; and C the declared task constraints. These include allowed grasp family, support/contact mode, target goal pose or region, uprightness, no-lift policy and immutable workspace geometry. The action definition must be sufficiently precise to evaluate success without referring to a hidden reference path.

The learner context is

$$c_0 = (o_0, \{p_i, G_i, \text{support}_i\}, p_t, G_t, q_0, g_0, m_0, W, a, g, C, \mathcal{D}).$$

Here g_0 denotes initial gripper aperture; the separate g denotes the task goal. $\mathcal{D}$ is the declared endpoint domain. Use explicit names in code (`gripper_gap`, `task_goal`) to avoid ambiguity.

Allowed learner input	Required provenance
RGB-D and instance masks	Initially GT masks in simulation; real masks are estimates
Object geometry, current pose and support	Supplied primitive geometry and explicit pose source
Initial robot joints, gripper and attachment state	Measured or simulator state; robot profile pinned
Fixed workspace and candidate resting poses	Identical information for matched baselines
Action, target identity, goal and constraints	Structured fields, not a hidden trajectory identifier

Excluded from the primary learner. Reference trajectories; sampled envelopes; target-path waypoint tokens; collision distances/contacts from an oracle; direct-blocker masks; target-planner results; successful teacher destinations; construction ancestry; planner seeds; root/family IDs; proof bounds; intervention outcomes. IDs identify slots in storage and output but are not ordinal geometry or embedding shortcuts. Clean images must exclude diagnostic overlays.

The model scores categorical endpoint assignments $x = (x_1, \dots, x_N)$. For obstacle i, $x_i = 0$ means stay at its actual initial pose; $x_i = k > 0$ references a candidate physical spot and pose. Define $M(x) = \{i : x_i \neq 0\}$ and $b(x) = |M(x)|$. No object is deleted. The target itself is not a repair obstacle in this benchmark.

Output contract. Object IDs and destinations, assignment score, moved count and optional abstention. The external executor determines relocation order and continuous robot motions. Do not claim that endpoint prediction alone produces an executable sequence.

2. Three separate validity questions

2.1 Static endpoint validity

For an assignment x, replace each obstacle pose with its selected pose while the target remains at its pre-action pose. Check support, fixed workspace, target-at-rest occupancy and pairwise obstacle compatibility:

$$Y_{\text{static}}(x) = \prod_i y_i^w(x_i) \prod_{i<j} y_{ij}^p(x_i, x_j).$$

These geometric factors do **not** assert that a target path exists. A valid endpoint can be inaccessible to the robot. A relocated object may legitimately occupy a region vacated by the robot; do not reject it merely because the initial arm occupied that region. Intermediate robot collisions belong to motion verification.

2.2 Executable repair and subsequent target action

Let $\Pi(x; S_0, C)$ be the set of legal move-once relocation sequences realizing x, including order, grasps, robot motions, achieved endpoint tolerances and all contact rules. For a complete sequence π, let $S_\pi = \text{Exec}(S_0, \pi)$. Let $\Gamma(S_\pi, a, g, C)$ be the allowed complete target motions from that achieved state. The research target is

$$Y^*(S_0, x, a, g, C) = \mathbf{1}[\exists \pi \in \Pi(x; S_0, C),\ \exists \gamma \in \Gamma(S_\pi, a, g, C) : \text{Valid}(\pi, \gamma) = 1]\,.$$

Robot start state, gripper state and attachment continuity are part of this definition. Different relocation orders can end at different robot states. A target witness from a substituted scene with an arbitrary home configuration is not automatically a witness for Y^*.

The former factorization is no longer exact. In general, existence of a path cannot be represented as a conjunction of per-object clearance labels. For example, either of two passages can suffice, but blocking both can prevent passage. Static pair compatibility does not capture that connectivity. The revised model therefore includes a whole-assignment term; it does not rename the old frozen-sweep unary as action feasibility.

2.3 Operational evaluation

The executable system uses a finite relocation executor and a finite target planner, not exhaustive existential search. An episode succeeds only when its chosen order, actual repair and target evaluation succeed. A witnessed $Y^* = 1$ is existence evidence; it does not guarantee a different finite executor will rediscover the witness.

Record these separately: `static_status`, `relocation_status`, `target_status`, `witness_status`, `simulation_status`, and `hardware_status`. An executable relocation followed by target-planning timeout is an operational failure-to-solve within budget, with epistemic status `UNRESOLVED`, not a certified $Y^* = 0$.

3. Evidence, minimum moves and the role of swept geometry

Use three-valued evidence at every applicable scope:

Evidence	Meaning
`VERIFIED_FEASIBLE`	A complete checked witness exists at the stated scope
`CERTIFIED_INFEASIBLE`	A sound proof excludes all choices in the explicitly declared scope
`UNRESOLVED`	Available search/checking did not decide the query

`NOT_RUN`, `ABORTED` and `CONSTRUCTION_UNAVAILABLE` describe execution/availability, not logical truth values. `OUT_OF_SCOPE` is an eligibility flag. Never turn a numerical error, incomplete IK search, finite RRT failure, or one rejected order into a global impossibility label.

Swept geometry policy

The primary learned proposal has no swept-volume input. The geometric backend may still calculate swept intersections, continuous collision checks or conservative interval bounds. Dataset constructors may use a frozen reference motion to find interesting perturbations. Those reference fields stay in `oracle_only` and do not define new primary action-success labels.

A frozen-sweep overlap is evidence about that motion only. A frozen collision is not evidence that all alternative target motions fail. A nominal trajectory or a reference feasible scene is not required at primary inference.

Minimum moves

$$b^* = \min_{x:Y^*(S_0,x,a,g,C)=1} |M(x)|.$$

Unless a stronger sound bound is established, use lower bound 0. A complete feasible repair-plus-target witness gives an upper bound. `MINIMUM_CERTIFIED` requires matching lower and upper bounds for this same objective, domain, contact model and allowed actions. A witnessed zero-move success proves minimum zero. Failure to find a zero-move path does not prove minimum at least one.

One useful sound lower-bound case is a fixed, unique target goal pose occupied by an immobile-unless-relocated obstacle: if the target and that obstacle must geometrically overlap at the required final goal, the obstacle must move. Prove the geometric premise and allowed-action scope; a goal region or alternative target orientation can invalidate the argument. A one-move complete witness then establishes a scoped optimum of one.

G3 theorem scope. The old result $b^*_{\text{terminal,frozen}} = |D_{\gamma_0}|$ still holds under its original identical-disc, vacant-reference-cell assumptions. It is not a lower bound for the new objective. The old A0 and Hall certificates remain valid only for their published fixed-motion endpoint domains. They must never be relabelled as arbitrary-action certificates.

Necessary static constraints may supply new sound bounds or Hall proofs, but every excluded edge must be justified without frozen-route clearance. Ordinary all-supported static matching often has the all-stay solution and gives bound zero.

4. Geometry, coordinates and complete robot checking

Retain the tested G1-G3 robot profile and its hash. Do not reset the robot to the provisional numeric profile printed in Guide v1.6. Current values must be read from the actual repository. This document has not inspected that repository.

Use meters, radians, float64 geometric calculations and float32 network tensors. Serialize 3D quaternions as `wxyz`. In 2D, world x points right and y backward along the support; array rows increase with world $+y$. In 3D, z points upward and the cupboard front is $y = 0$. State any image-row reversal explicitly.

For the initial 8×8 lattice, pitch $h = 0.08$ m and cell (r, c) has centre $((c + 1/2)h, (r + 1/2)h)$. The footprint radius is 0.024 m; clearance margin is 0.005 m. Thus distinct cells satisfy $0.08 > 2(0.024) + 0.005 = 0.053$ m. Cell uniqueness exactly captures obstacle-pair nonoverlap only for the declared identical footprints and spacing. Mixed sizes or overlapping off-grid candidates require explicit pair constraints.

The grid limits resting endpoints, not intermediate movement. Objects and all robot links follow continuous paths between cells. One active target is excluded from the movable-obstacle set; other targets remain collision bodies.

Full articulated 2D contract

For base b, link lengths L_j and cumulative angles $\theta_j = \sum_{k \le j} q_k$,

$$p_{\mathrm{TCP}}(q) = b + \sum_{j=1}^{3} L_j(\cos\theta_j, \sin\theta_j), \qquad \phi_{\mathrm{TCP}} = q_1 + q_2 + q_3.$$

Check both declared IK branches, joint limits, finite-width links, housings, palm, two fingers, carried object, support/fixed geometry, and nonadjacent self-collision. Preserve the tested layered-link collision convention and document its physical scope. The G3 report uses 48 declared grasp candidates; do not silently change this to the older master document's 64.

Relocation phases remain pregrasp, approach, closure/attachment, carry, release/detachment and retreat, with continuity from the previous achieved state. The target action starts from the final achieved robot state. **No connector to a historical γ_0 start is required in the new mode.** Ordinary approach/connector motions within a newly planned skill still require checking.

Continuous verification

Apply clearance once to uninflated geometry. With signed distance d, demand $d \ge \delta$, and retain a declared numerical gray band. Sampling collision-free nodes is not a continuous certificate. Use analytic checks, a validated continuous routine, or conservative adaptive subdivision with an established distance-change bound. If an interval cannot be decided within its budget, record `UNRESOLVED`. Validate any simulator distance API against contact, penetration and near-miss cases; bounded distance results are not unlimited exact distances.

5. Endpoint model and learned whole-assignment energy

Retain frozen visual features pooled by visible instance mask and trainable object projections. For images I_v and mask weights w_{ivp},

$$\bar{f}_i = \frac{\sum_{v,p} w_{ivp} F_\phi(I_v)_p}{\sum_{v,p} w_{ivp}}, \qquad z_i = f_\theta(\text{stopgrad}\, \bar{f}_i).$$

Missing visible area is a missing-observation condition, not an observed zero feature. Known primitive shape and explicit poses remain supplied inputs. Do not claim perception of unknown shape or perfect real-world poses.

Candidate token $h_i(k; c_0)$ combines appearance, known dimensions, initial and proposed pose, displacement, support, target-relative and goal-relative geometry, and robot/action context. Include fixed workspace representation and constraint encoding. Recompute proposed geometry for each candidate; ghost placement substitutes the same object, never clones it.

Static factors

Shared unary and symmetric pair networks predict $y_i^w(k)$ and $y_{ij}^p(k,l)$. Use exchange-symmetric pair processing, e.g. average logits under swapping arguments. Convert logits into nonnegative energies with $e = \text{softplus}(-\ell)$. These are static compatibility energies only. Keep factors for stay choices and all present objects.

Whole-assignment success head

A compact initial architecture uses shared token MLPs, permutation-invariant sum pooling and a final MLP:

$$u(x, c_0) = \rho_\theta\left(\sum_i \phi_\theta(h_i(x_i; c_0)), N, \text{context}(c_0)\right),$$
$$\hat{p}_{\text{task}}(x \mid c_0) = \sigma(u), \qquad e_{\text{task}}(x, c_0) = \text{softplus}(-u).$$

Pooling width starts at 128; use two-layer MLPs. This is an implementable starting architecture, not an expressivity or performance guarantee. It can represent assignment-level interactions that a product of independent action-clear factors cannot. An attention architecture is optional only after a measured limitation and matched comparison.

The complete score is

$$E_\theta(x \mid c_0) = \sum_i e_i^w(x_i) + \sum_{i<j} e_{ij}^p(x_i, x_j) + \lambda_{\text{task}} e_{\text{task}}(x, c_0).$$

The task head predicts prospects for a complete repair-plus-target witness under declared constraints, using only initial and proposed scene information. It receives no planned path. It is neither a proof nor a physical success probability without separate calibration. Local and global evidence may be correlated, so the total energy must not be presented as an exact factorization of true success probability.

6. Boltzmann proposals and assignment selection

Keep categorical endpoints and an at-most-b distinct-move budget:

$$P_{\theta,b}(x \mid c_0) = \frac{\mathbf{1}[|M(x)| \le b] \exp[-E_\theta(x \mid c_0)/T]}{Z_{\theta,b}(c_0)}, \qquad T > 0.$$

The network weights are fixed at inference. Gibbs sampling changes hypothetical endpoint assignments, not the observed scene and not model weights. Randomize object-update order, use seeded independent chains, include all-stay and mixed legal initializations, deduplicate candidates and record unique proposal counts.

The old local-only Gibbs conditional is insufficient. For candidate value k of object i, the conditional includes the assignment-level term:

$$H_i(k; x_{-i}) = e_i^w(k) + \sum_{j \neq i} e_{ij}^p(k, x_j) + \lambda_{\text{task}} e_{\text{task}}(x_{-i}, k; c_0).$$

Use ordered pair arguments consistently for $j < i$. Sample proportional to $\exp[-H_i/T]$ over values respecting the move budget. Constants independent of k cancel. For sum pooling, cache the sum of object embeddings, subtract the old contribution and add each alternative; recompute the nonlinear final task head for each candidate. Validate this against full energy recomputation.

A learned global term removes the old purely local factor-table complexity claim. Report setup, candidate scoring, sampler and verifier time separately. Finite chains are proposals, not guaranteed equilibrium samples or complete search.

Primary selection policy

Generate a fixed pool across budgets including zero, using only c_0. Apply the same declared static screen to matched methods. Among candidates above a task-score threshold τ chosen on validation data, prioritize fewer moved objects, then higher score; deduplicate and freeze this ranking. Threshold and budgets are configuration, never tuned on test target outcomes.

The external relocation planner may test this ranked list and different orders until it finds an executable repair or exhausts its declared budget. It must not call the target planner while choosing a repair. The first accepted relocation witness is committed. Target evaluation is run afterward, once per primary episode, with a predeclared internal planner budget. A failed target evaluation cannot advance to the next proposal in that episode.

All-stay is a valid output; the empty relocation is immediately executable. Its task score still matters, otherwise a static-only filter would choose zero moves for every scene. If no candidate passes the threshold, return `ABSTAIN`; retain the episode in success denominators. Report threshold sweeps and coverage on validation, and one frozen operating point on test.

7. Offline labels and training losses

Target planning is allowed during offline teacher generation and labeling. It is prohibited as hidden online proposal feedback in the primary protocol. Store a mechanically enforced separation between learner observations and evidence.

Label hierarchy

Label	Positive evidence	Negative evidence
Workspace/support factor	Decisive geometric check	Decisive geometric violation
Endpoint pair factor	Decisive nonoverlap	Decisive overlap/occupancy violation
Complete assignment Y^*	Full relocation-plus-target witness from S_0	Sound impossibility certificate for all allowed orders/motions realizing this assignment
One attempted sequence	Verified sequence and target	Definite violation of that sequence only; not an assignment-wide negative

A static-invalid assignment is a certified complete-assignment negative because any realization of those required endpoints violates the rules. A unique goal pose still occupied by a stationary obstacle can provide an additional action-level negative under a checked geometric certificate. Finite planner misses remain unknown. Report the negative-source mix; easy invalid-endpoint negatives alone do not demonstrate learning of action feasibility.

Store `witness_found_with_budget` as a separate operational field if useful. Do not train the main feasibility head with timeout-as-zero while calling it existential feasibility. An optional planner-budget predictor would have different labels, claims and evaluation, and is outside the initial architecture.

Losses

Use balanced factor-type sampling for

$$\mathcal{L}_{\text{static}} = \text{mean}_{f \text{ decided}} \, \text{BCEWithLogits}(\ell_f, y_f).$$

For assignments with a full witness or a sound negative certificate,

$$\mathcal{L}_{\text{task}} = \text{mean}_{x \text{ decided}} \, \text{BCEWithLogits}(u(x, c_0), Y^*(x)).$$

Unresolved examples are stored but masked from these binary losses. For a deduplicated pool C_{dec} of decided complete-assignment labels with positive subset C^+, define

$$\widetilde{P}(x) = \frac{e^{-E(x)/T_{\text{train}}}}{\sum_{v \in C_{\text{dec}}} e^{-E(v)/T_{\text{train}}}}, \quad q_C(x) = \frac{\mathbf{1}[x \in C^+]}{|C^+|},$$

$$\mathcal{L}_J = -\sum_{x \in C_{\text{dec}}} q_C(x) \log \widetilde{P}(x), \quad \mathcal{L} = \mathcal{L}_{\text{static}} + \lambda_s \mathcal{L}_{\text{task}} + \lambda_J \mathcal{L}_J.$$

Use the joint term only for mixed positive/negative pools. Exclude unknowns from both numerator targets and denominator; inclusion as competitors would implicitly penalize unknown feasible repairs. The derivative with respect to an independent energy is $(q_C - \widetilde{P})/T_{\text{train}}$. Test this sign numerically. The uniform-positive target is a finite-pool training convention, not a distribution over all feasible repairs.

If witness or certified-negative coverage is insufficient, G4 reports a data-readiness failure. Do not invent negatives or proceed to scientific training claims.

8. Intervention evidence and attribution

The proposal identifies repair participants and destinations. A chosen repair object is not automatically a uniquely necessary causal blocker. Different valid routes can admit different sufficient repairs, and an indirectly moved object may enable grasp access rather than the target path itself.

Offline, construct matched interventions by changing a documented subset of object poses while restoring all other physical state. Keep the action, goal, robot profile, constraints and evaluation protocol fixed. Target trajectories are allowed to differ across these evaluations; the question is action feasibility, not a fixed motion's collision response.

For a legal intervention I, the conceptual action-feasibility effect is

$$\Delta_I = Y_{\text{action}}(S_0^I, a, g, C) - Y_{\text{action}}(S_0, a, g, C).$$

Only report a definite binary effect when both sides have decided evidence. A planner timeout on the original scene and a witness after intervention support **observed recovery within the protocol**, not a proof of a $0 \to 1$ physical feasibility change. Repeat paired planner seeds where stochastic operational differences are measured, and state the finite scope.

Distinguish diagnostic pose substitution from physically executable relocation. Offline teleportation can generate controlled scenes but cannot serve as a relocation witness. Removing an object is a different intervention from moving it; deletion is not an allowed repair in the primary task.

Claims and evidence

Claim	Required support
Sufficient repair	Verified complete repair and target action
Participant in the returned repair	Object is actually moved in that witness
Necessary member of this repair	All relevant omission alternatives excluded within stated scope
Minimum number of moved objects	Sound action-level lower bound equals complete witness count
Intervention-trained model helps	Matched training/evaluation ablation with equal coverage and label budgets
Latent causal identifiability	Not claimed by this project

Successful alternative repairs are all valid labels, even if they move different object sets. Do not use a single teacher set as the unique ground-truth culprit mask. Repair-set overlap can be reported as teacher agreement; it is not the primary correctness metric. Complete action success and move count take priority.

For matched scene pairs, keep rendering seeds, nuisance factors and object identities controlled where possible. Export a changed-field audit. Provenance, family names and correction histories remain outside the learner input. Generate dedicated no-change and nuisance-only controls to detect memorization of rendering or construction artifacts.

9. Execution protocol and component boundaries

The primary protocol is `INITIAL_SCENE_ONE_SHOT_V1`. Its pseudocode is:

```
PROPOSE(c0, weights, seed):
    score hypothetical endpoint assignments using only c0
    sample/deduplicate a fixed candidate pool including all-stay
    screen static geometry; rank by frozen selection rule
    return ranked assignments or ABSTAIN

COMMIT_REPAIR(S0, ranked_assignments, repair_budget):
    for x in ranked_assignments:
        plan move-once order and relocations from S0
        if a full relocation witness is verified:
            return committed(x, order, relocation_witness)
    return NO_REPAIR_WITNESS_WITHIN_BUDGET

EXECUTE_AND_EVALUATE(S0, committed):
    execute the committed repair; track achieved states and H
    if deviation or execution failure invalidates it: ABORT
    after all relocations, call the target planner/evaluator
    validate/execute the target motion from achieved state
    record success, unresolved, or scoped impossibility
    do not feed target outcome back to proposal selection
```

For offline benchmarking, execute/replay the committed repair on a cloned state and evaluate the target from its resulting achieved state. On hardware, evaluate from the actual post-repair state. Target evaluation must not precede repair completion or filter/replace the committed choice in the primary metric. A separate practical preflight/retry variant may use such feedback, but is named and reported separately.

For hardware execution, reobserve after each relocation to check pose, gripper and attachment tolerances. Low-level replanning toward the same committed next destination is permitted if logged and budgeted; it cannot change object selection/destinations or invoke the target planner. Abort when assumptions fail. A secondary closed-loop proposal method may update assignments, but is not the one-shot method.

History H counts physically moved object identities, not hypothetical sampler changes or failed planning attempts. Every selected obstacle moves at most once in an episode. Preserve history across controller retries and observations. Candidate planning occurs on clones of S_0; failed candidate search must not mutate the live scene. Do not teleport back after a failed physical move to try another nominally one-shot repair.

The target planner receives the final achieved joints, gripper state, object poses and full constraints. It plans the complete skill and checks goal achievement, not merely endpoint IK. No original reference trajectory or connector-to-reference requirement is present.

10. Dataset, case families and split contract

Every primary record contains a clean initial scene, structured action/goal, finite candidate maps, initial robot state, static factor labels, complete-assignment evidence, proposed/achieved repairs, target outcomes, scope and costs. Store initial and final robot states separately. Label generation uses the same allowed physical actions as evaluation.

Separate namespaces are mandatory:

Namespace	Contents
`learner`	Allowlisted initial observations, geometry, robot/action context and candidates
`labels`	Decided static and complete-assignment labels, masks and certificate references
`oracle_only`	Reference paths, construction edits, planner traces, direct blockers, witnesses
`execution`	Committed assignment, ordered achieved states, moved history and outcomes

Root IDs are metadata for splitting only. Split before intervention descendants, alternate goals, alternative trajectories, render variants or hard-negative mining. Keep all siblings in one root split. Do not use a filename or planner hash as an implicit model feature.

New development cases

Case	Required observation and scientific purpose
R0: already feasible	Complete zero-move target witness; test all-stay and unnecessary moves
R1: frozen path blocked, action feasible	Same initial scene defeats a stored reference but has a different zero-move witness; separates objectives
R2: goal occupancy	Unique required goal overlaps one obstacle; certify the occupancy premise and demonstrate one-move recovery
R3: alternative sufficient repairs	At least two different assignments have full witnesses; do not assume either is minimal
R4: order/access dependence	Same assignment has a rejected/unresolved order and a verified order, all from the same initial state
R5: static-valid, target failure/unknown	Endpoints pass; target has a sound obstruction certificate or remains unresolved, with the distinction explicit
R6: invalid and unknown controls	Duplicate destination, unsupported endpoint, narrow numerical band, forced planner exhaustion and provenance leakage checks

Constrained hinge/drawer sweeps are a later extension, not an automatic G3.5 requirement. Do not build a new articulated-object simulator just to complete the migration case table.

Begin with small full-robot cases and archive G3 reconstructions as comparison controls. Use dense 2D scenes with 30-40 obstacles only after the new interface works. Density and move count are measured properties; do not demand five necessary blockers using the old construction. The 3D pilot remains 6-10 obstacles.

Scale after the gate: 20 audited roots, then 300 independent roots (210/45/45), then optional 1,000-root expansion if throughput and evidence coverage justify it. Report roots, sibling queries, candidate assignments, decided labels and unique execution witnesses separately. Preserve unknown records and their computational costs.

11. Evaluation and baseline protocol

The primary evaluator never supplies a target result to the learned selection module. Compare methods using the same input geometry, candidate domains, collision backend, relocation executor and target planner budget. Report both fixed-call and observed wall-time costs where practical.

ID	Method	Interpretation
B0	No relocation; target planner only	Measures how often repair was unnecessary
B1	Static geometry + deterministic assignment search	Nonlearned, target-blind repair proposal control; no exact action-optimum claim
B2	Seeded uniform legal assignment proposals	Proposal-structure control with declared budget policy
B3	Geometry-only local factors + assignment head	Tests learned action-level repair without appearance
B4	Visual + geometry local factors + assignment head	Primary learned model
B5	Static local/pair energy only	Tests the contribution of whole-assignment action conditioning
B6	Same trained energy, greedy vs Gibbs	Tests proposal search while holding representation fixed
B7	Retrained model without task goal/action context	Diagnostic ambiguity ablation; not expected to solve general queries
B8	Matched observational vs intervention training	Tests controlled interventions at equal roots/labels/coverage
B9	Global/local training vs additional finite-pool joint loss	Tests joint objective rather than mere task-head presence
O1	Offline target-oracle-assisted search	Privileged reference/upper-performance comparator; excluded from equal-information claims
F0	Archived frozen-motion A0 and learned controls	Different objective; results in a separate panel

For B1/B2/B5, declare the move-budget/selection policy in advance and compare success-move curves; do not give them target feedback to select budgets. For B4 also test no visual features, pair lesion and retrained pair removal when making corresponding claims. Keep the earlier A0-A11 registry historical; do not silently reuse an A-label with new semantics.

Primary metrics: complete episode success; relocation success; target success conditional on completed relocations; moved count; success versus move budget; abstention/coverage; unnecessary moves on independently witnessed initially feasible cases; proposal, IK, relocation-planner and target-evaluation calls; runtime breakdown; certificate/unknown rates; execution deviations. Count abstentions, planning exhaustion and execution aborts in the overall denominator. Report out-of-scope exclusions separately.

Define N as all eligible evaluated queries. Overall success is n_{complete}/N. Target-after-repair success is $n_{\text{complete}}/n_{\text{repair completed}}$, with all-stay counted as completed empty repair and zero denominator reported N/A. Report a separate repair-only conditional metric if desired. Never infer initially blocked cases from a baseline timeout: distinguish certified initially infeasible, witnessed initially feasible and initially unresolved cohorts.

Bootstrap over independent roots. Minimum-move accuracy is reported only for the subset with sound action-level minima. Report a verified witness count as an upper bound elsewhere. A learned method beating weak static controls does not establish advantage over oracle-assisted planning without accounting for the information and cost difference.

12. 3D, real execution and long-horizon integration

The 3D benchmark remains the full pinned xArm7, gripper and cupboard, with upright cylinders and support-contact planar transport. This is an embodiment change, not a new target objective. Inputs remain trajectory-free, and the target planner runs from the achieved state after repair commitment.

For cylinder height 0.075 m and support top 0.005 m, nominal centre height is 0.0425 m. Require throughout each verified motion:

$$|z_{\text{obj}}(q(t)) - 0.0425| \le \epsilon_z, \qquad \angle(R_{\text{obj}}(q(t))e_z, e_z) \le \epsilon_{\text{tilt}}.$$

Robot links are free to move in 3D. Object yaw may change under the declared side-grasp policy; no vertical overflight is allowed. Candidate paths must respect these constraints between nodes, not only at IK endpoints.

Body pair	Contact policy
Selected object and support	Intended support contact throughout supported transport
Selected object and fingers	Allowed only during validated grasp/attachment phases
Selected object and other obstacles	Forbidden
Robot and cupboard/walls/roof	Forbidden
Robot self pairs	Check nonadjacent pairs; document allowed exclusions

Pin asset revisions, meshes, joint conventions, TCP, tool-to-object transform, mount, home, clearance, allowed contacts, controllers and assistance flags. `UNSET` assets block 3D results. A simulation weld is attachment assistance, never evidence of a physically stable grasp. Kinematic supported transport does not establish friction, force tracking or slip robustness.

Evidence ladder

`STATIC_VALID` is endpoint geometry only. `PLANAR_ARM_VERIFIED` and `FULL_ROBOT_VERIFIED` require complete checked relocation and target trajectories. `SIM_REPLAY_SUCCESS` additionally requires declared dynamics/controller execution. `HARDWARE_SUCCESS` requires measured real execution. Each level has separate fields; no automatic promotion.

Real inputs require calibrated camera-to-robot transforms, object/target pose estimates, tracked identities and known or estimated geometry. Log uncertainty and achieved pose deviations. A simulator margin is not automatically adequate for hardware. Validate clearance and grasp/controller behaviour for the actual apparatus before claiming transfer. No new hardware operation is authorized by this document.

Long horizon

G10 uses a scripted sequence of at least three target actions. At each action boundary, obtain a fresh initial scene for a new one-shot repair query. Within each episode preserve its fixed proposal and moved history. Reset history only after success or explicit abandonment. Show a case where a locally successful repair interferes with a later action; report local and complete-script success separately. The initial model has no plan-suffix input and makes no global task-optimality claim.

13. Migration from G3 and acceptance gates

G3.5 is an implementation and evidence audit, not training. Its detailed execution instructions are in `G3_5_initial_scene_repair_migration.md` and Guide v1.7.

1. Inspect the repository, instruction history, actual profile/configuration and archived G3 evidence. Record commit and scope. Do not assume this PDF has already validated the code.
2. Keep the old mode callable with unchanged archived paths and labels. Add an explicit new protocol flag and independent action/goal hash. Frozen trajectory identity is not a required primary query field.
3. Separate static endpoint checking, relocation verification and post-commit target evaluation. Target evaluation must be instrumentably unreachable during primary proposal and relocation-candidate selection.
4. Use hand-authored or deterministic candidates for R0-R6. They test the contract, not learned intelligence. Demonstrate full witnesses where available and preserve unresolved outcomes.
5. Remove the frozen target-start connector requirement only in the new mode. The target planner must receive the actual end state of the chosen relocation order.
6. Define and audit allowlisted learner tensors, label statuses and certificate scope. Store reference sweeps as optional privileged diagnostics only.
7. Validate new tests and applicable legacy replays, then write report 0007 with the case matrix, actual visualizations, measured costs and a G4 readiness decision.

Do not scale data or train at this gate. If a requested case cannot be constructed with the actual robot, record `CONSTRUCTION_UNAVAILABLE` with evidence; do not invent a witness or substitute a disc-only drawing. A failed required gate blocks the dependent stage rather than causing fabricated completion.

Release invariants

Invariant	Acceptance evidence
No trajectory information reaches primary model	Allowlist and poisoned privileged-field test
No target feedback selects the repair	Spy/call trace: zero target calls before commit; no next candidate after target result
Initial scene defines the proposal	Proposal hash and source-observation hash stable through execution
Endpoint validity is not action validity	R1/R5 controls and separate labels
Whole-assignment energy is used correctly	Global conditional agrees with full energy differences
Unknowns are not binary negatives	Loss masks and finite-budget failure test
Target starts at achieved state	State-continuity assertions and replay
Old minima remain scoped	Frozen report unchanged; new results have separate bounds

14. Limitations and exact change ledger

This is a finite-endpoint, move-once, action-conditioned research benchmark. It assumes known primitive geometry, explicit pose provenance and an eligible initial robot state. It does not prove global TAMP completeness, recover identifiable latent causal variables, discover hidden objects, solve arbitrary dynamics, or guarantee neural repair correctness. It does not yet demonstrate useful one-shot repair; G3.5/G4 must establish that enough meaningful supervision can be generated.

The global success head is an architectural proposal necessitated by the broader label semantics. Its accuracy, calibration, generalization and benefit over geometric search remain experimental questions. Known identical cylinders may make visual features redundant; the geometry-only baseline is mandatory. Invalid-endpoint negatives can make training look successful without learning connectivity or access. Report hard-negative evidence and end-to-end outcomes explicitly.

v1.4 area	v1.5 replacement
Scientific question and immutable decisions	Initial-scene, action/goal-conditioned repair; new protocol name
Frozen action/envelope	Optional teacher/control artifact; excluded from primary input
Action-clear unary label y_i^a	Removed from primary factor semantics
Exact target-validity product	Restricted to static endpoint factors; existential action success defined separately
Energy	Added whole-assignment task head and its full Gibbs contribution
Losses	Added witness/certificate assignment labels and task BCE; unknowns excluded from joint normalization
Inference target checks	Removed before and between moves; one post-commit target evaluation
Reobservation	Execution validation and abort; no hidden one-shot proposal refresh
A0 matching/minimality	Frozen control retained; no action-level optimality implied
Causal attribution	Sufficient repairs, alternative sets and certificate-scoped necessity
Dataset	New R0-R6 controls, namespace leakage checks and evidence coverage gate
Evaluation	New B/O/F registry; honest initial-status cohorts and component timings
3D and long horizon	Same revised objective; preserved physical and evidence constraints
Stage registry	Insert G3.5; revise G4-G10 without erasing G0-G3

Source basis

This revision was written from the supplied specification v1.4, implementation guide v1.6, G3 completion report 0006 dated 16 October 2026, and the user's clarification in this conversation. The older master document supplies historical motivation only. No repository files, experiments, trained models or hardware were changed by this document-writing task.

Next action: implement and review G3.5 using the companion instruction file. G4 may start only after its data and interface prerequisites pass and the user authorizes the next stage.